

Numbers and Maths

JavaScript has only one number type -> Number

Number :-

integers -> 10, -5, 100, -5000

floats -> 3.14, 0.1, 0.3

Special value :- infinity, - infinity, NaN(not a number)

Arithmetic operator:-

operator :-

+ addition 5 + 5 = 10

- subtraction 5 - 5 = 0

* multiplication 5 * 5 = 25

/ division 5/5 = 1

% mod 5 % 2 = 1

** exponent 2 ** 3 = 8

Type conversion:-

// console.log(typeof "10"); --> 10

// console.log(Number("abc")); --> NaN

Order of operations and Brackets:-

BODMAS/PEMDAS:-

PEMDAS:- PARENTHESES() , EXPONENTS **, MULTIPLICATION, DIVISION, ADDITION , SUBTRACTION

BODMAS :- BRACKET, OPERATOR, DIVISION, MULTIPLICATION, ADDITION, SUBTRACTION

1) PARENTHESES()

2) EXPONENTS **

3) MULTIPLICATION/DIVISION

4) ADDITION AND SUBTRACTION

When the order of operation has same priority, your math is solved from left to right.

Floating point inaccuracy:-

```
console.log(0.1 + 0.2); // 0.300000000004
//This happens because computers store number in binary.
// some decimal numbers cannot be represented exactly
```

solution:-

use integers:-

```
(10 + 20)/100 --> 0.3
```

-- Some important inbuilt Math methods

1) Math.round():-

Math.round() rounds off your floating point number to the closest Integer number.

```
Math.round(123.34 + 345.45); --> 469
```

2) Math.floor():-

Math.floor() rounds off your floating point number to the previous closest integer number.

```
console.log(Math.floor(1.99)); --> 1
```

3) Math.ceil():-

Math.ceil() rounds off your floating point number to the next closest integer number.

```
console.log(Math.ceil(1.01)); --> 2
```

